THE LIFTING CATEGORY IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 963-967 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 969-1016, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_stacks-stacks-more-morphisms.json (pb=2246, stmt 963-967, proof 969-1016).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: the detailed strong-cartesian proof is replaced by the essential reason (unique factorization mechanism) - artin #30 precedent.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02XK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1016–1016. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02XM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1016–1016. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 003V: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1016–1016. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-morphism: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 919–928. Cited by the retained

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statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CF4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 711–747. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-gerbe-of-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 875–903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07VW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1016–1016. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The category $p : \mathcal{C} \to W_{spaces,étale}$ constructed in Remark remark-gerbe-of-lifts is fibred in groupoids.*

Proof. We claim the fibre categories of p are groupoids. If (f, f', γ') as in (equation-morphism) is a morphism such that $f : U \to V$ is an isomorphism, then f' is an isomorphism by Lemma 0CF4 and hence (f, f', γ') is an isomorphism.

Consider a morphism $f : V \to U$ in $W_{spaces,étale}$ and an object $\xi = (U, U', a, i, x', \alpha)$ of $\mathcal{C}$ over U. We are going to construct the "pullback" $f^*\xi$ over V. Namely, set $b = a \circ f$. Let $f' : V' \to U'$ be the étale morphism whose restriction to V is f (More on Morphisms of Spaces, Lemma 07VW). Denote $j : V \to V'$ the corresponding thickening. Let $y' = x' \circ f'$ and $\gamma = \mathrm{id} : x' \circ f' \to y'$. Set

$$\beta = \alpha \star \mathrm{id}_f : x \circ b = x \circ a \circ f \to x' \circ i \circ f = x' \circ f' \circ j = y' \circ j$$

It is clear that $(f, f', \gamma) : (V, V', b, j, y', \beta) \to (U, U', a, i, x', \alpha)$ is a morphism as in (equation-morphism). The morphisms (f, f', γ) so constructed are strongly cartesian (Categories, Definition 02XK). We omit the detailed proof, but essentially the reason is that given a morphism $(g, g', \epsilon) : (Y, Y', c, k, z', \delta) \to (U, U', a, i, x', \alpha)$ in $\mathcal{C}$ such that g factors as $g = f \circ h$ for some $h : Y \to V$, then we get a unique factorization $g' = f' \circ h'$ from More on Morphisms of Spaces, Lemma 07VW and after that one can produce the necessary ζ such that $(h, h', \zeta) : (Y, Y', c, k, z', \delta) \to (V, V', b, j, y', \beta)$ is a morphism of $\mathcal{C}$ with $(g, g', \epsilon) = (f, f', \gamma) \circ (h, h', \zeta)$.

Therefore $p : \mathcal{C} \to W_{étale}$ is a fibred category (Categories, Definition 02XM). Combined with the fact that the fibre categories are groupoids seen above we conclude that $p : \mathcal{C} \to W_{étale}$ is fibred in groupoids by Categories, Lemma 003V. □